Outlier Rejection

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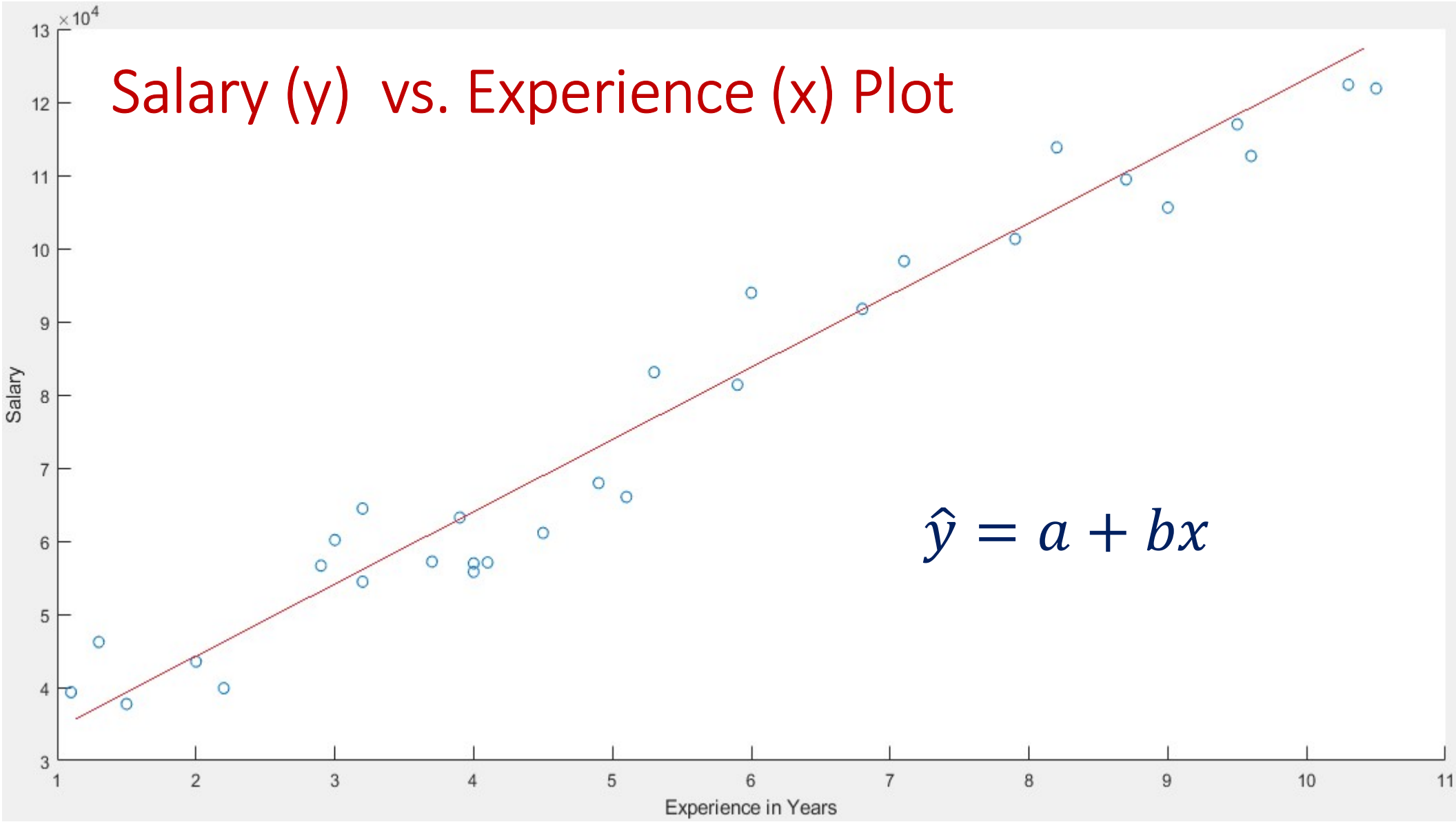
Overview

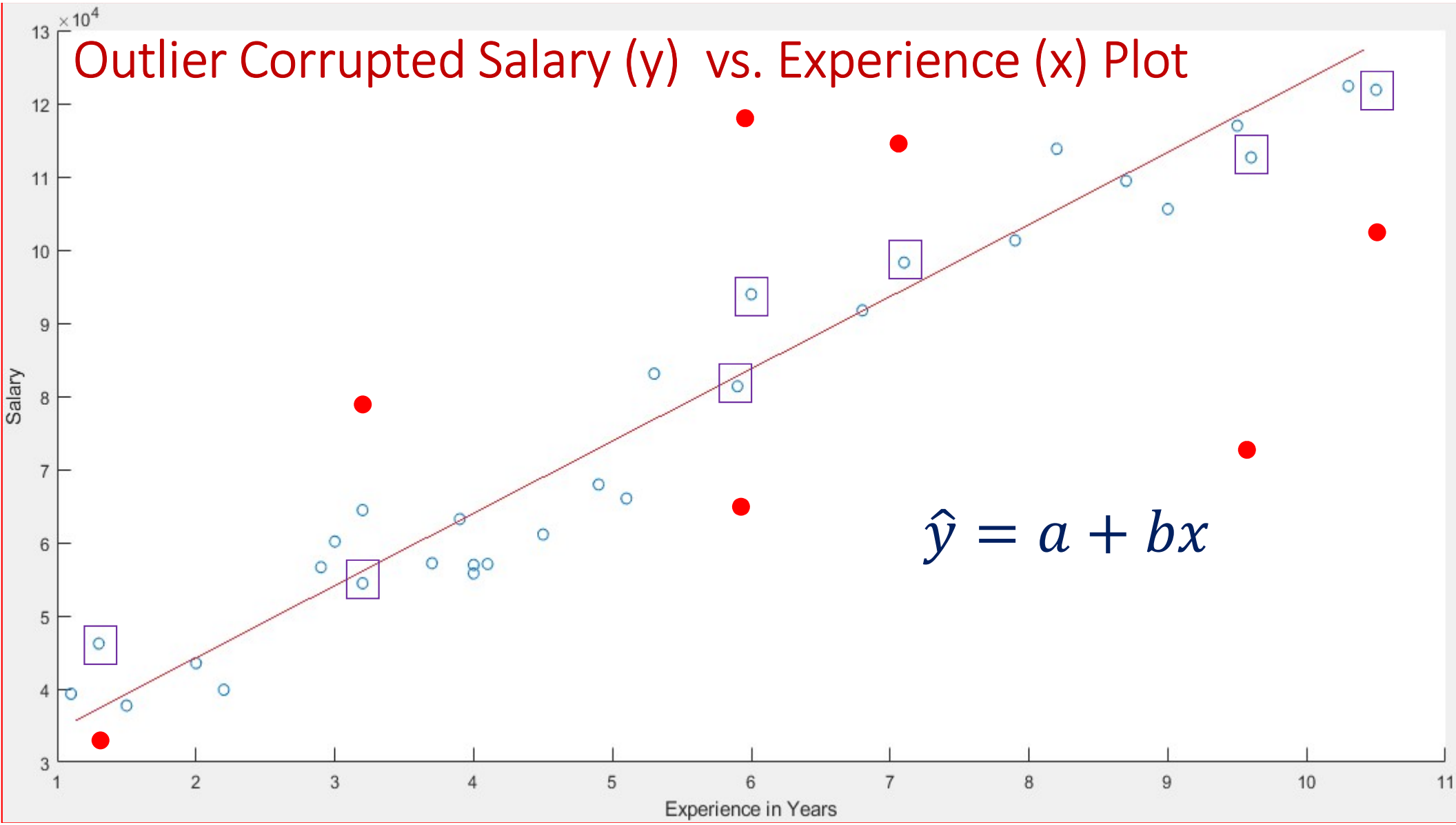
- Recapitulation - Linear Regression Problem
- Concept of Pseudo Inverse
- Recapitulation - Chebyshev Inequality
- Outlier Rejection Problem
- Regression Diagnostics & IRLS Formulation
- RANSAC - RANdom SAmple Consensus

Salary (y) vs. Experience (x) Plot



$$\hat{y} = a + bx$$

Outlier Corrupted Salary (y) vs. Experience (x) Plot



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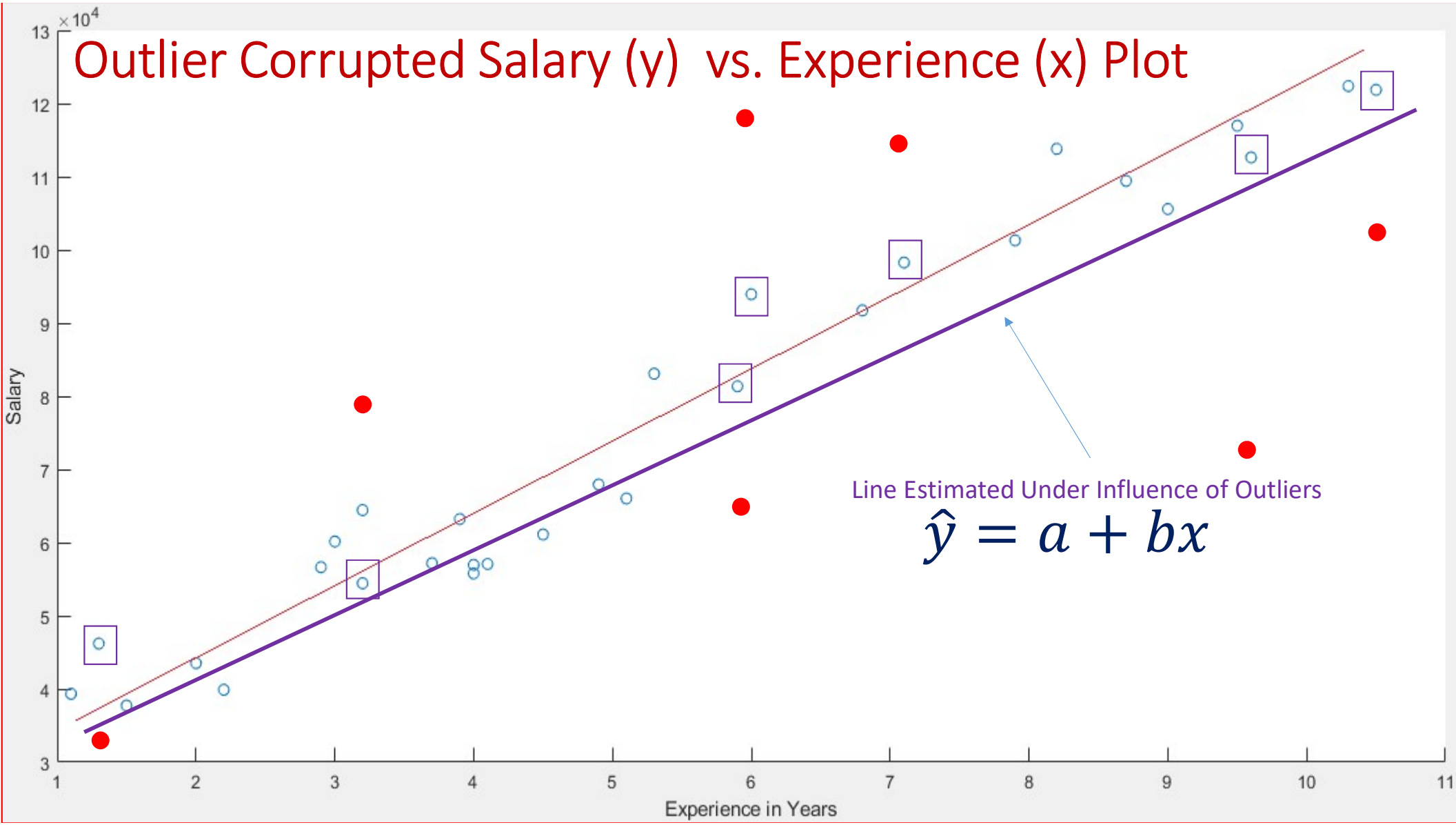


The Linear Regression Problem

$$y = a + \sum_{k=1}^{m} b_k x_k$$

Linear Regression Problems involving m Independent Variables

$$\left\{\left(x_1^{(i)}, \dots x_m^{(i)}, y^{(i)}\right); i = 1, \dots n\right\}$$

The dataset containing n training instances

$$e_i = y^{(i)} - \left\{a + \left(b_1 x_1^{(i)} + \cdots + b_m\, x_m^{(i)}\right)\right\}$$

Linear Model Prediction Error for Instance i

Linear Regression: The Error Vector

$$E = \begin{bmatrix} e_1 \\ \vdots \\ e_i \\ \vdots \\ e_n \end{bmatrix} = \underbrace{\begin{bmatrix} y^{(1)} \\ \vdots \\ y^{(i)} \\ \vdots \\ y^{(n)} \end{bmatrix}}_{\mathbf{Y}} - \underbrace{\begin{bmatrix} 1 & x_1^{(1)} & \cdots & x_m^{(1)} \\ \vdots & \vdots & \cdots & \vdots \\ 1 & x_1^{(i)} & \cdots & x_m^{(i)} \\ \vdots & \vdots & \cdots & \vdots \\ 1 & x_1^{(n)} & \cdots & x_m^{(n)} \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} a \\ b_1 \\ \vdots \\ b_m \end{bmatrix}}_{\mathbf{P}}$$

Linear Regression: Solution by Pseudo-Inverse

$$\mathrm{E_T} = E^T E = \sum_{i=1}^{n} e_i^2$$

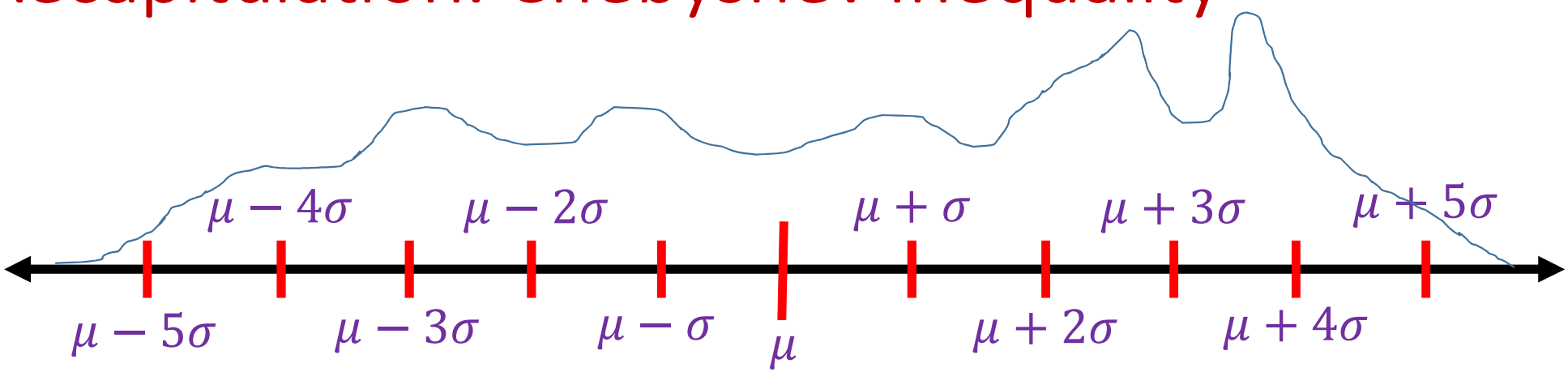
The Total Error

Minimize $\mathrm{E_T} = E^T E$ w.r.t. P

$$P = (A^T A)^{-1} A^T Y = A^{\#} Y$$

Pseudo-Inverse of A: $A^{\#} = (A^T A)^{-1} A^T$

Recapitulation: Chebyshev Inequality

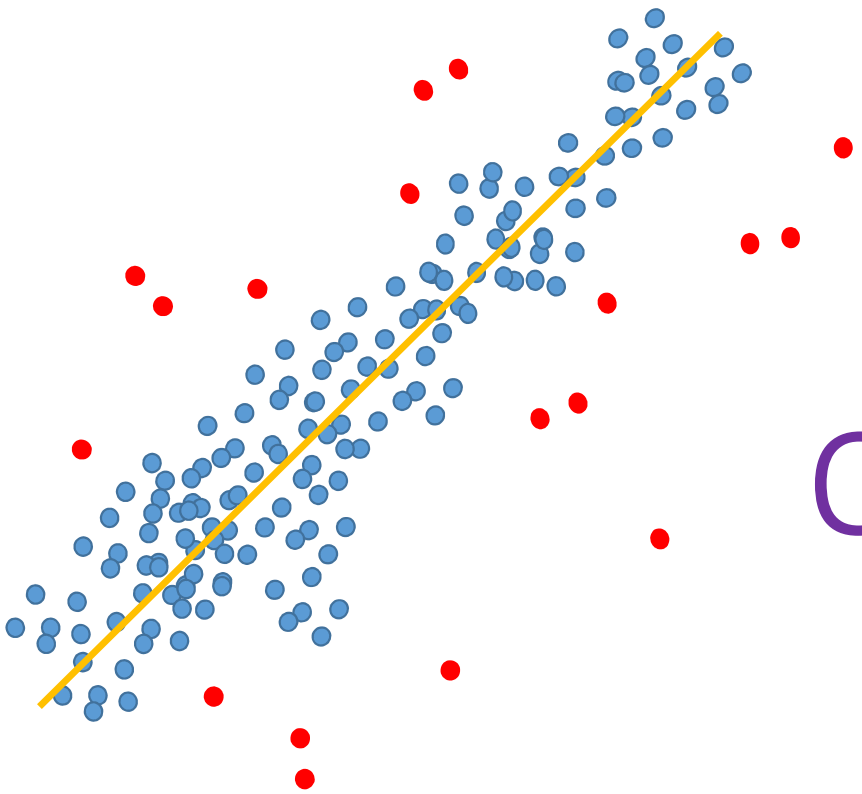


$$P\{|X - \mu| \geq \lambda\sigma\} \leq \frac{1}{\lambda^2}$$

Recapitulation: Chebyshev Inequality

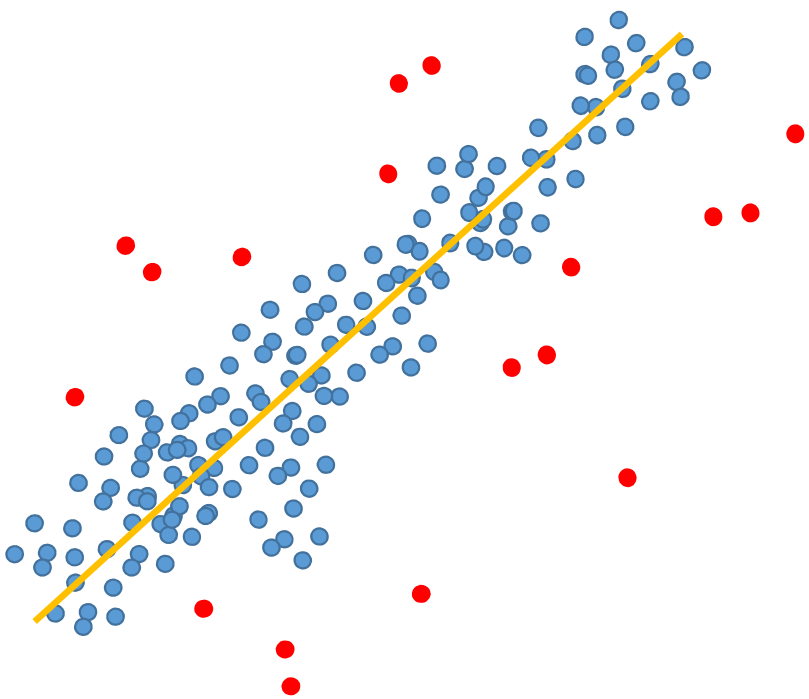
Provides an Upper Bound on Membership for an Instance based on its Distance from Mean Normalized by Standard Deviation. No Properties of the Underlying Distribution are Assumed.

$$P\{|X - \mu| \geq \lambda\sigma\} \leq \frac{1}{\lambda^2}$$



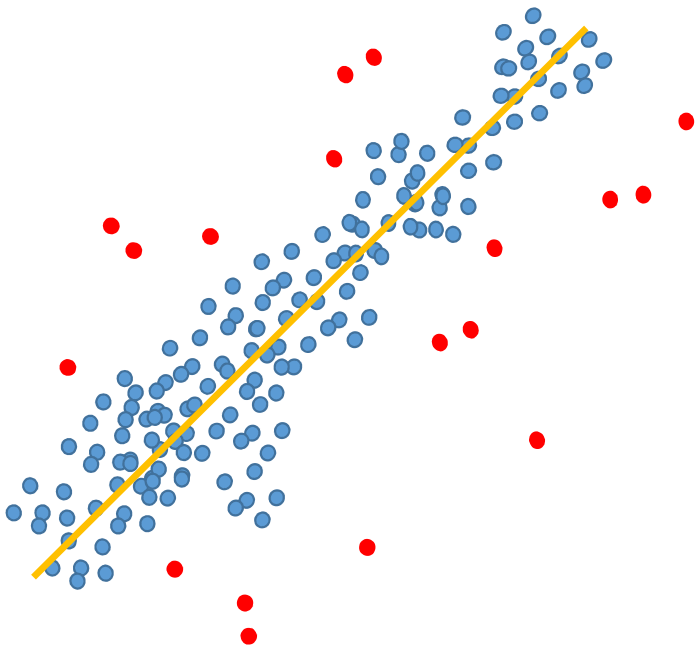
Outlier Rejection

Outlier Rejection in Linear Regression



- Original Linear Regression
  - Model (y=a+bx) is known
  - To Estimate (a,b)
  - From input $(x_i, y_i); i = 1, \dots n$
- Consider Data Corruption by Outliers
  - Points that adhere to Model (BLUE)
  - Outliers (RED)
- Probability Guided Predictions
  - Regression Diagnostics
  - RANSAC

Regression Diagnostics



- Input: Outlier Corrupted Dataset
- Fit *Model* on the Dataset
- Evaluate the *Model* on the Dataset
- Check Distribution of *Prediction Errors*
- *Outliers* Lead to *Large Prediction Errors*
- *Trim* Dataset by *Removing Outliers*
- Iterate till *Termination*
- Termination on Dataset Size Limit

Regression Diagnostics: Simplest Linear Regression

- Consider Points $\mathrm{D} = \{(x_i, y_i); i = 1, \dots n\}$ and Set $\mathrm{D}_1 = \mathrm{D}$
- Iteration #k: Sets/Parameters are Subscripted/Superscripted by (k)
- Fit a line $\mathrm{y} = \mathrm{a}_\mathrm{k}^\star + \mathrm{b}_\mathrm{k}^\star x$ with parameters $(\mathrm{a}_\mathrm{k}^\star, \mathrm{b}_\mathrm{k}^\star)$ using input dataset D_k
- Compute Error $\mathrm{e}_\mathrm{i}^{(\mathrm{k})} = \mathrm{y}_\mathrm{i}^{(\mathrm{k})} - \left(\mathrm{a}_\mathrm{k}^\star + \mathrm{b}_\mathrm{k}^\star x_i^{(k)}\right); \left(x_i^{(k)}, y_i^{(k)}\right) \in D_k;\ i = 1, \dots |D_k|$
- Compute Second Order Statistics from $\mathrm{e}_\mathrm{i}^{(\mathrm{k})}; i = 1, \dots |D_k|$
- Define: $\mu_k = \frac{1}{|D_k|} \sum_{i=1}^{|D_k|} |e_i^{(k)}|$ and $\sigma_k^2 = \frac{1}{|D_k|} \sum_{i=1}^{|D_k|} \left(|e_i^{(k)}| - \mu_k\right)^2$
- Construct Set $\mathrm{D}_\mathrm{k}^\mathrm{O} = \left\{\left(x_j^{(k)}, y_j^{(k)}\right) : \left|e_j^{(k)}\right| \geq \mu_k + \lambda\sigma_k\right\}$
- Outliers Rejected in Iteration k: $\mathrm{D}_\mathrm{k}^\mathrm{O}$
- Construct More Pure Set of Inliers for Next Iteration as $\mathrm{D}_{\mathrm{k}+1} = \mathrm{D}_\mathrm{k} - \mathrm{D}_\mathrm{k}^\mathrm{O}$
- Terminate as $\frac{|D_{k+1}|}{|D|} \leq \eta$ OR $k = K_{max}$

Regression Diagnostics: General Regression Problems

- Consider Points $\mathrm{D} = \{(\boldsymbol{x}_i, \boldsymbol{y}_i); i = 1, \dots n\}$ and Set $\mathrm{D}_1 = \mathrm{D}$
- Iteration #k: Sets/Parameters are Subscripted/Superscripted by (k)
- Fit a model $\mathbf{y} = h(\boldsymbol{x}; \boldsymbol{\theta}_k^\star)$ with parameters $\boldsymbol{\theta}_k$ using input dataset D_k
- Compute Error $\mathrm{e}_\mathrm{i}^{(\mathrm{k})} = \left\| \boldsymbol{y}_i^{(k)} - h\left(\boldsymbol{x}_i^{(k)}; \boldsymbol{\theta}_k^\star\right) \right\|_2 ; \left(\boldsymbol{x}_i^{(k)}, \boldsymbol{y}_i^{(k)}\right) \in D_k;\ i = 1, \dots |D_k|$
- Compute Second Order Statistics from $\mathrm{e}_\mathrm{i}^{(\mathrm{k})}; i = 1, \dots |D_k|$
- Define: $\mu_k = \frac{1}{|D_k|} \sum_{i=1}^{|D_k|} |e_i^{(k)}|$ and $\sigma_k^2 = \frac{1}{|D_k|} \sum_{i=1}^{|D_k|} \left(|e_i^{(k)}| - \mu_k\right)^2$
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- Outliers Rejected in Iteration k : D_k^O
- Construct More Pure Set of Inliers for Next Iteration as $D_{k+1} = D_k - D_k^O$
- Terminate as $\frac{|D_{k+1}|}{|D|} \leq \eta$ OR $k = K_{max}$

Iteratively Reweighted Least Squares (IRLS)

Initialization

- Consider Points $\mathrm{D} = \{(\boldsymbol{x}_i, \boldsymbol{y}_i); i = 1, \dots n\}$
- Iteration #k: Parameters are Sub-/Superscripted by (k)
- Set Initial Weights $\omega_i(1) = \frac{1}{n}\ ; i = 1, \dots n$

Model Fitting

- Define Instance Error: $e_{ik} = \|\boldsymbol{y}_i - h(\boldsymbol{x}_i; \boldsymbol{\theta}_k)\|_2$
- Define Total Weighted Error: $E_k = \sum_{i=1}^{n} \omega_i(k-1) e_{ik}^2$
- Minimize E_k to find Optimal Model Parameters $\boldsymbol{\theta}_k^\star$

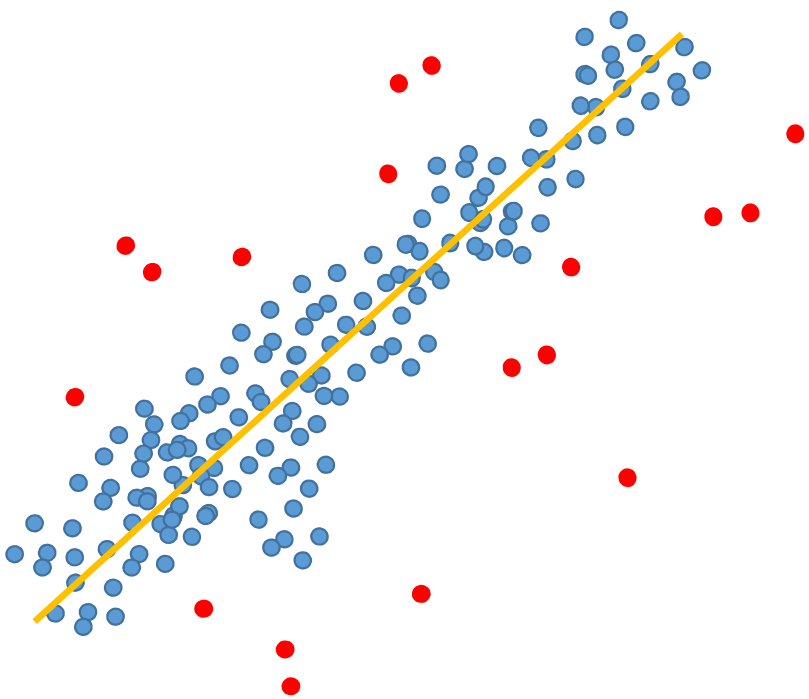
Weight Update

- Define Instance Error for Optimal Parameter: $\epsilon_{ik} = \|\boldsymbol{y}_i - h(\boldsymbol{x}_i; \boldsymbol{\theta}_k^\star)\|_2$
- Compute Parameters: $v_i(k) = \frac{1}{1+\gamma\epsilon_{ik}^2}$
- Update Weight: $\omega_i(k) = \frac{v_i(k)}{\sum_{j=1}^{n} v_j(k)}$

Iterations

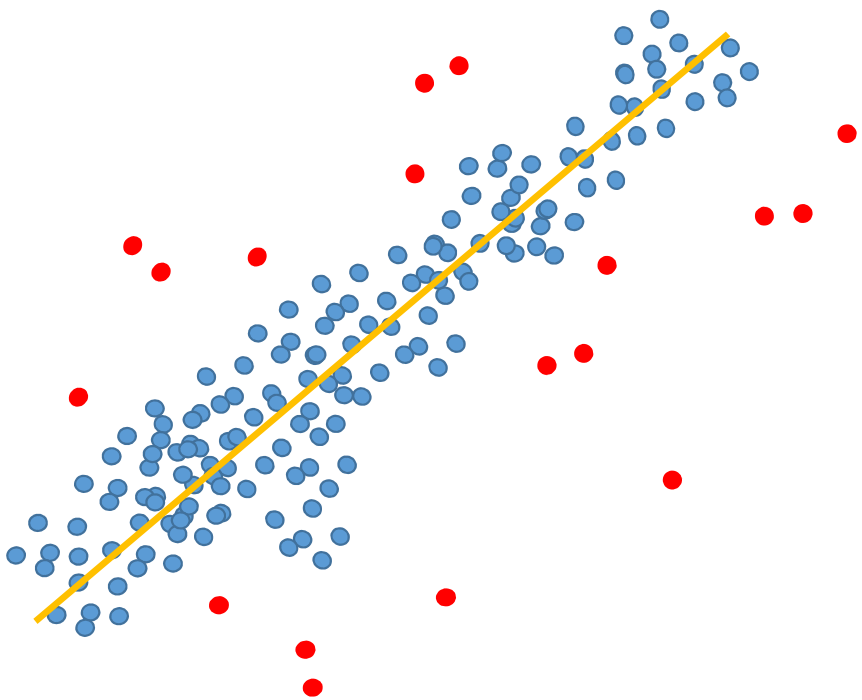
Termination: By Convergence of Parameters OR By Maximum Number of Iterations

RANSAC



- RANdom SAmpling Consensus
- Regression Diagnostics Iteratively Rejects Outliers
- RANSAC Performs Multiple Trials on Small Data Subsets
- Each Trial Draws a Small Dataset for Model Fitting
- All these Models coming from Multiple Trials are Tested
- The one with Minimum Error is Accepted

RANSAC: Definitions



- Total Number of Points: n
- Total Number of Inliers: n_i
- Total Number of Outliers: n_o
- Fraction of Inliers: γ

$$n = n_o + n_i$$

$$\gamma = \frac{n_i}{n}$$

RANSAC: The Trial K

- Each Draw Needs *m* Points
- Successful Draw has All *m* Inliers
- Success at the Trial *k* (say)
- Earlier *(k-1)* Trials were unsuccessful
- Define Event *A*: All Points Drawn are Inliers
- Define Event B_k: Success at Trial *k*

$$P(A) = \gamma^m = b\ (say)$$

$$P(\check{A}) = 1 - \gamma^m = a\ (say)$$

$$P(B_k) = \gamma^m (1 - \gamma^m)^{k-1} = ba^{k-1}$$

RANSAC: Number of Trials

Expected Number of Trials using Mathematical Expectation

Evaluate Mean and Variance of the Distribution

$$E[k] = \sum_{k=1}^{\infty} kba^{k-1}$$

$$Var(k) = E[(k - E[k])^2]$$

Use Identity

$$1 + a + a^2 + a^3 + \cdots \infty = \frac{1}{1-a} = \sum_{k=0}^{\infty} a^k$$

$$1 + 2a + 3a^2 + 4a^3 + \cdots \infty = \frac{1}{(1-a)^2} = \sum_{k=1}^{\infty} k a^{k-1}$$

$$1.2 + 2.3.\, a^1 + 3.4.\, a^2 + 4.5.\, a^3 + \cdots \infty = \frac{2}{(1-a)^3} = \sum_{k=2}^{\infty} k(k-1) a^{k-2}$$

RANSAC: Expected Number of Trials

$$E[k] = \sum_{k=1}^{\infty} bka^{k-1} = b\sum_{k=1}^{\infty} ka^{k-1}$$

$$E[k] = \frac{b}{(1-a)^2} = \frac{1}{b}$$

RANSAC: Variance of the Number of Trials

$$E[\{k - E(k)\}^2] = E[k^2 - 2kE(k) + \{E(k)\}^2]$$

$$= E\left[k^2 - \frac{2k}{b} + \frac{1}{b^2}\right] = E[k^2] - \frac{2}{b}E[k] + \frac{1}{b^2}$$

$$= E[k^2] - \frac{2}{b} \times \frac{1}{b} + \frac{1}{b^2} = E[k^2] - \frac{1}{b^2}$$

RANSAC: Variance of the Number of Trials

$$E[k^2] - \frac{1}{b^2} = \sum_{k=1}^{\infty} k^2 b a^{k-1} - \frac{1}{b^2}$$

$$= \sum_{k=1}^{\infty} (k^2 - k) b a^{k-1} + \sum_{k=1}^{\infty} k b a^{k-1} - \frac{1}{b^2} = \sum_{k=1}^{\infty} k(k-1) b a^{k-1} + \frac{1}{b} - \frac{1}{b^2}$$

$$= ab \sum_{k=1}^{\infty} k(k-1) a^{k-2} + \frac{b-1}{b^2} = \frac{2ab}{(1-a)^3} + \frac{b-1}{b^2} = \frac{2(1-b)b}{b^3} + \frac{b-1}{b^2}$$

$$= \frac{1-b}{b^2}$$

RANSAC: Expected Number of Trials

- Expected Number of Trials using Mathematical Expectation
- Evaluate Mean and Variance of the Distribution

$$E[k] = \sum_{k=1}^{\infty} kba^{k-1} = \frac{1}{b} = \frac{1}{\gamma^m}$$

$$E[(k - E[k])^2] \sim \frac{1}{b^2}$$

Chebyshev Inequality Threshold

$$T_{total} > \frac{\lambda + 1}{\gamma^m}$$

Summary

- Recapitulation - Linear Regression Problem
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- Recapitulation - Markov & Chebyshev Inequalities
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- Regression Diagnostics
- RANSAC - RANdom SAmple Consensus

Thank You